

CENGİZ TOPEL, Turkey

WMO: 172005

Lat: 40.735N

Lon: 30.083E

Elev: 182

StdP: 14.60

Time Zone: 3.00 (E03)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	23.0	26.4	21.2	16.0	28.1	24.4	18.7	31.8	19.6	58.9	17.5	54.8	1.6	320	0.423

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.9	91.8	72.1	89.5	71.8	87.4	71.4	76.4	85.0	75.1	83.6	73.8	82.3	8.5	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.7	126.5	79.9	73.0	123.2	79.5	71.4	116.7	78.4	40.0	84.9	38.8	84.1	37.4	81.6	82.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.9	14.3	12.3	20.0	98.2	4.1	4.2	17.0	101.2	14.6	103.7	12.3	106.1	9.3	109.2		
			18.8	78.8	4.2	2.3	15.7	80.4	13.2	81.7	10.8	83.0	7.8	84.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	58.2	42.1	43.8	48.0	54.2	62.8	70.0	74.0	75.3	68.3	60.6	52.5	45.2
	DBStd	13.14	7.87	8.03	7.20	6.33	5.53	4.66	3.84	4.16	4.90	6.41	6.84	7.83
	HDD50	870	267	206	123	26	1	0	0	0	0	5	49	193
	HDD65	3459	708	594	527	330	110	8	0	0	24	165	378	615
	CDD50	3846	24	32	63	152	397	600	744	785	548	332	124	45
	CDD65	961	0	1	2	6	41	158	280	320	122	27	3	1
	CDH74	9769	1	4	22	127	538	1649	2881	3148	1108	249	37	5
	CDH80	3452	0	0	2	24	131	555	1083	1271	326	57	3	0
(14) Wind	WSAvg	4.5	4.2	4.8	5.2	5.0	4.6	4.8	5.1	5.0	4.4	3.8	3.6	3.7
(15) Precipitation	PrecAvg	31.4	3.5	3.0	2.9	2.3	2.0	2.5	1.4	1.6	2.1	3.2	2.9	4.1
	PrecMax	43.0	6.6	6.6	4.7	5.6	5.7	8.2	4.5	7.9	5.4	7.7	5.7	9.4
	PrecMin	21.7	0.9	0.7	1.2	0.2	0.2	0.1	0.0	0.0	0.0	0.4	0.6	1.7
	PrecStd	5.5	1.5	1.5	0.9	1.2	1.3	1.9	1.3	1.7	1.5	1.9	1.5	1.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.4	71.9	77.1	82.8	87.9	93.6	95.2	96.4	91.8	85.6	78.9	73.2
		MCWB	53.5	55.8	56.7	62.7	68.3	72.7	73.2	72.3	69.3	65.1	61.8	56.1
	2%	DB	64.1	66.4	71.7	77.4	83.9	88.2	91.4	92.5	87.5	80.5	73.2	66.9
		MCWB	52.1	52.5	54.8	60.6	67.3	70.6	72.9	72.4	69.3	65.2	59.9	54.4
	5%	DB	59.8	62.7	67.6	73.2	80.3	86.0	88.1	89.5	83.8	75.6	68.4	62.7
		MCWB	50.0	51.5	53.8	58.7	65.7	70.3	71.7	73.0	68.5	64.1	58.3	52.9
	10%	DB	55.5	57.5	62.7	68.2	76.6	82.7	86.0	86.4	80.5	71.8	64.4	58.6
		MCWB	48.4	49.7	52.0	57.1	64.5	69.5	71.3	72.3	68.0	63.2	56.8	51.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.5	57.7	58.8	65.0	72.0	76.4	78.0	78.3	75.2	69.9	65.3	58.5
		MCDB	63.3	68.6	71.9	77.9	81.3	86.8	87.4	87.0	82.2	75.6	72.8	67.4
	2%	WB	52.9	54.3	56.4	62.1	69.5	74.2	75.8	76.8	72.6	68.1	62.5	56.0
		MCDB	61.7	63.5	67.4	73.7	79.1	83.5	84.7	85.3	80.4	73.7	68.4	64.1
	5%	WB	50.7	52.1	54.7	59.8	67.5	72.5	74.6	75.6	70.8	66.1	59.9	54.0
		MCDB	57.5	60.6	64.9	70.5	76.3	81.7	83.5	83.8	79.1	73.4	65.9	61.0
	10%	WB	49.0	50.3	52.8	57.8	65.6	70.8	73.2	74.4	69.3	64.3	57.6	51.9
		MCDB	54.8	57.7	61.7	67.1	74.0	79.5	81.8	82.5	77.2	70.7	63.5	57.2
(35) Mean Daily Temperature Range	5% DB	MDBR	16.0	16.8	20.0	22.3	23.1	23.4	23.6	22.9	22.9	20.2	19.1	17.4
		MCDBR	23.4	25.5	30.8	32.5	30.7	28.6	27.8	27.0	30.3	27.3	26.3	23.7
		MCWBR	13.1	14.1	15.8	17.2	15.7	13.6	12.3	10.8	14.0	14.0	14.8	13.2
	5% WB	MCDBR	20.8	23.2	25.6	28.1	25.7	24.7	23.4	22.4	23.4	20.9	21.9	20.2
		MCWBR	12.1	13.4	14.1	15.6	14.5	12.8	11.5	10.0	11.9	11.3	13.4	11.8
(40) Clear-Sky Solar Irradiance	taub		0.359	0.388	0.430	0.471	0.441	0.426	0.430	0.436	0.423	0.403	0.369	0.359
	taud		2.326	2.229	2.127	2.036	2.196	2.284	2.289	2.269	2.291	2.322	2.362	2.348
	Ebn at Noon		247	256	258	255	267	270	268	263	258	249	242	236
	Edh at Noon		29	36	44	52	45	42	41	41	37	33	27	26
(44) All-Sky Solar Radiation	RadAvg		493	730	1107	1513	1896	2169	2247	1965	1473	976	658	461
	RadStd		70	104	106	172	172	190	86	129	117	87	84	68

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	+0.86	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-162	N/A	N/A

Nomenclature: See separate page